Category Theory for Mathematical Physics — Exercise Sheet 1

Categories: a theory of systems and processes

Variant: Questions only

Conventions as in Lecture 1: in $g \circ f$ the morphism f acts first; BG has one object $*$ with $\mathrm{Hom}(*,*) = G$ and $g \circ h = gh$; $\mathrm{Rep}(G)$ has the finite-dimensional complex representations and all intertwiners. A *homomorphism* of groups preserves products (and hence the identity). The cyclic group $\mathbb{Z}_n$ is written additively. In Exercise 3, $\omega = e^{2\pi i/3}$.

Exercise 1 (Categories, non-examples, and groups). (i) Take as objects the natural numbers $0, 1, 2, \ldots$, with exactly one morphism $m \to n$ when $m \leq n$ and none otherwise. Show that this is a category, and find its isomorphisms.

(ii) Take as objects the finite-dimensional complex vector spaces, and as morphisms $V \to W$ the linear maps of rank at most 1, composed as maps. Show that a composite of two such maps is again one. Show that this is not a category, and find the objects V that have an identity morphism.

(iii) Show that a functor $BG \to BH$ is exactly a homomorphism $G \to H$. How many functors are there $B\mathbb{Z}_4 \to B\mathbb{Z}_2$, and how many $B\mathbb{Z}_2 \to B\mathbb{Z}_3$?

(iv) Let $\varphi, \psi \colon G \to H$ be homomorphisms, read as functors $BG \to BH$. Show that a natural transformation $\varphi \Rightarrow \psi$ is an element $h \in H$ with $\psi(g) = h\,\varphi(g)\,h^{-1}$ for all $g \in G$, and that it is always a natural isomorphism. There are 10 homomorphisms $S_3 \to S_3$; how many classes do they form up to natural isomorphism?

Exercise 2 (G-sets are functors $BG \to \mathrm{Set}$). A *G-set* is a set X with an action $g \cdot x$, such that $1 \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$; a map $f \colon X \to Y$ of G-sets is *equivariant* if $f(g \cdot x) = g \cdot f(x)$.

(i) Show that a functor $BG \to \mathrm{Set}$ is a G-set, and that a natural transformation between two such functors is an equivariant map.

(ii) Let $G = S_3$, let $X = \{1, 2, 3\}$ with the permutation action, and let $Y = \{+, -\}$ on which the even permutations act trivially and the odd ones swap $+$ and $-$. Count the equivariant maps $X \to X$, $X \to Y$, $Y \to X$ and $Y \to Y$.

(iii) For a finite G-set X, let $\mathbb{C}[X]$ have basis e_x ($x \in X$) and $\rho(g)e_x = e_{g \cdot x}$. Show that $\mathbb{C}[X]$ is a representation and that an equivariant $f \colon X \to Y$ gives an intertwiner $\mathbb{C}[f] \colon e_x \mapsto e_{f(x)}$, and that $X \mapsto \mathbb{C}[X]$, $f \mapsto \mathbb{C}[f]$ is a functor from finite G-sets to $\mathrm{Rep}(G)$.

Exercise 3 (Restriction is a functor, faithful and not full). Let $H \leq G$ be a subgroup and $\iota \colon H \to G$ the inclusion.

(i) Show that restriction, $\mathrm{Res}(V, \rho) = (V, \rho \circ \iota)$ and $\mathrm{Res}(T) = T$, is a functor $\mathrm{Rep}(G) \to \mathrm{Rep}(H)$, and that on objects, read through Theorem 1.9, it is composition with the functor $B\iota \colon BH \to BG$.

(ii) Show that Res is faithful.

(iii) Let $G = S_3$, $H = A_3$, and $V = \{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ with the permutation action, the standard representation. Show that $\dim \mathrm{Hom}_{S_3}(V, V) = 1$ and $\dim \mathrm{Hom}_{A_3}(\mathrm{Res}\, V, \mathrm{Res}\, V) = 2$. Conclude that Res is not full.

(iv) Write down an intertwiner of $\mathrm{Res}\, V$ that does not come from $\mathrm{Hom}_{S_3}(V, V)$.

Exercise 4 (The dual over the reals). Let $\mathrm{Vect}_{\mathbb{R}}^{\cong}$ be the groupoid of finite-dimensional real vector spaces and linear isomorphisms, with $D(V) = V^*$ and $D(f) = (f^*)^{-1}$ as in Lecture 1.

(i) Show that no family of linear isomorphisms $\theta_V : V \to V^*$ is a natural transformation $\mathrm{id} \Rightarrow D$ on $\mathrm{Vect}_{\mathbb{R}}^{\cong}$.

(ii) On the groupoid of finite-dimensional real inner-product spaces and orthogonal maps, show that $R_V(u) = \langle u, \cdot \rangle$ is a natural isomorphism $\mathrm{id} \Rightarrow D$. Why does this contradict neither (i) nor Theorem 1.11?

(iii) For which real numbers λ does the test $f = \lambda\, \mathrm{id}_V$ prove (i)? What plays the role of λ in the lecture's proof for unitary maps, and why is there no such scalar among the orthogonal maps?

Exercise 5 (Skeletons). (i) Let FinBij be the groupoid of finite sets and bijections, and S the category with objects $0, 1, 2, \ldots$, $\mathrm{Hom}(n, n) = S_n$ and no morphisms $m \to n$ for $m \neq n$. Show that S is a skeleton of FinBij, and that the two are not isomorphic.

(ii) Let $\mathcal{G}$ be a groupoid in which any two objects are isomorphic, and x an object. Show that $\mathcal{G} \simeq B\,\mathrm{Aut}(x)$, where $\mathrm{Aut}(x) = \mathrm{Hom}(x, x)$. Apply this to the groupoid of the two-element subsets of $\{1, 2, 3\}$ with bijections.

(iii) In the proof of Theorem 1.16 the bases were chosen with $b_{\mathbb{C}^n} = \mathrm{id}$. Keep every choice but one: $b_{\mathbb{C}^2} = \mathrm{diag}(1, 2)$. Compute $G(M(A))$ for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Is $G \circ M = \mathrm{id}$? Are M and G still an equivalence?

(iv) Lecture 1's functor "forgetting crossings" goes from the braid groupoid Br to the category S of (i). Show that it is full and not faithful. You may use one fact about braids: two braids on the same strands are equal in Br only if they have the same *exponent sum*, the number of positive crossings minus the number of negative ones, where a crossing is positive when the strand going up to the right passes over.

Exercise 6 ⋆. *Natural transformations of the identity of* $\mathrm{Rep}(G)$. Let G be finite, $\mathbb{C}[G]$ its group algebra (basis $g \in G$, product extended linearly), and ρ_V extended linearly to $\mathbb{C}[G]$. A natural transformation $\xi\colon \mathrm{id}_{\mathrm{Rep}(G)} \Rightarrow \mathrm{id}_{\mathrm{Rep}(G)}$ is an intertwiner $\xi_V : V \to V$ for each representation V, with $T \circ \xi_V = \xi_W \circ T$ for every intertwiner $T : V \to W$.

(i) For z in the centre $Z(\mathbb{C}[G])$, show that $\xi_V = \rho_V(z)$ is such a natural transformation.

(ii) Conversely, let ξ be one, let $R = \mathbb{C}[G]$ be the regular representation, $\rho_R(g)a = ga$, and put $z = \xi_R(1)$. Draw the naturality of ξ for an intertwiner $T : R \to V$ as a box sliding through ξ, in the style of Lecture 1's Equation (1.2). Show that z is central and that $\xi_V = \rho_V(z)$ for every V.

(iii) Show that $z = \sum_g z_g g$ is central exactly when z_g is constant on conjugacy classes. Natural transformations of $\mathrm{id}_{\mathrm{Rep}(G)}$ are added and scaled component by component, and composed as in Lemma 1.8. Conclude that they form an algebra isomorphic to $Z(\mathbb{C}[G])$, of dimension the number of conjugacy classes, and give that number for S_3 and for the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$.